

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>
we know the file size is 200 gb so we can write that down
then we know carla downloads at a rate of 2 gb per minute so we can do the first part of the download
by doing the file size which is 200 divided by the download rate of 2
so we do 200 split by 2 which equals 100 so now we know the first part of the download taking 100
minutes
then the question tells us that 40 percent way through the download there's a restart so we take the 100
minutes we just worked out and do 40 percent of that
to work out 40 percent of 100 we do 100 split by 100 to get 1 then times that 1 by 40 which equals 40
so now we know 40 minutes into the download there's a restart
so now we know the first part of the download takes 100 minutes and then there's a 40 minute restart
so now we add those 100 and 40 together which equals 140
but then the question tells us that after the restart carla has to start the whole download again from
the beginning so we take our 140 minute amount and add the 100 minutes from the start of the
download again
so we take our 140 and add the 100 which equals 240
</think><answer>240</answer>